

Exploring Climate Change Effects on Sinkhole Formation: a Long-Term Temperature Analysis in Long Island, NY (1948-2024)



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Abstract

In recent years, there has been an overall increase in sinkhole occurrences in Long Island, NY. Sinkholes typically form when water erodes detrital or evaporite rock beneath the Earth's surface, creating a cavity that eventually collapses. Increased underground water mobility may lead to accelerated sinkhole development, especially upon more frequent warming temperatures due to climate change. We have analyzed historical atmospheric temperature data in Long Island, NY, spanning 76 years (1948-2024) by applying the Kolmogorov-Zurbenko (KZ) filter to investigate the number of prolonged freezing and thawing events per year. We have utilized R language coding (Rstudio) to account for the number of times that the temperature fluctuated at two specific temperature intervals of freezing (-4 to 1°C) and thawing (4 to 10°C). We have found that the number of prolonged thawing events has increased while the number of freezing times has decreased, leading to an increase in water mobility and, ultimately, the exacerbation of underground erosion. Understanding the relationship between temperature changes and water mobility is crucial for assessing the potential impacts of climate change on sinkhole dynamics and for implementing effective mitigation strategies in vulnerable regions of Long Island.

Introduction

- Our study investigates the mechanisms of sinkhole formation in Long Island's gravel-dominated surface lithology.
- The rate of sinkhole occurrences on Long Island has increased over the past decade or so
- There is limited research to suggest why sinkhole occurrences might be increasing, let alone a correlation between sinkhole formation and the effects of climate change.
- The current understanding of sinkhole formation is supported by water mobility; if water is able to move more underground, more erosion is possible.
- Another contributor toward physical erosion is freeze-thaw action, which can create and expand underground gaps, further enabling water mobility during times of thawing.

Methodology

- Data acquisition and cleaning:
 - Utilized RStudio with packages GSODR, KZA, and rNOAA
 - Gathered data directly from weather stations in the Tristate area
 - Interpolated small gaps using the imputeTS package
- Station selection: John F. Kennedy International Airport
- Analysis:
 - Script written in RStudio to count occurrences of temperature within specific intervals
 - Intervals: Freezing (-4°C to 1°C) and thawing (4°C to 10°C)
 - Applied a moving average using the KZ filter with an 11-year window in order to negate the effect of 11-year sunspot cycles (Tentomas & Marsellos 2022)
- KZ filter:
 - Robust performance in non-stationary data and outlier presence
 - Addresses abrupt changes or anomalous data points iteratively
 - Ensures stable and reliable output, avoiding signal loss or over-smoothing

Results

The results did ultimately support our hypothesis. A negative trend was observed in the total number of times temperatures crossed the freezing window per year as well as a positive trend in the total number of times temperatures crossed the thawing window per year (Fig 2). Additionally, a positive trend in average yearly temperature since 1948 was observed (Fig 1).

Discussion & Conclusion

- Aim: Explore correlation between global warming and sinkhole occurrence
- Focus: Relationship between rising global temperature, water mobility, and sinkhole formation
- Methodology: Use R language to gather historical temperature data from NOAA weather stations
- Comparison: Compare data with existing research findings
- Findings:
 - Support hypothesis of increased temperatures impacting sinkhole formations
 - Decline in freeze-thaw cycles over time
 - Ice melting and remaining unfrozen underground
 - Development of water mobility within Long Island soil
- Observation: Sinkhole formation due to climate change is recent in northern latitudes
- Historical context: Prolonged winter times in the past 60-80 years

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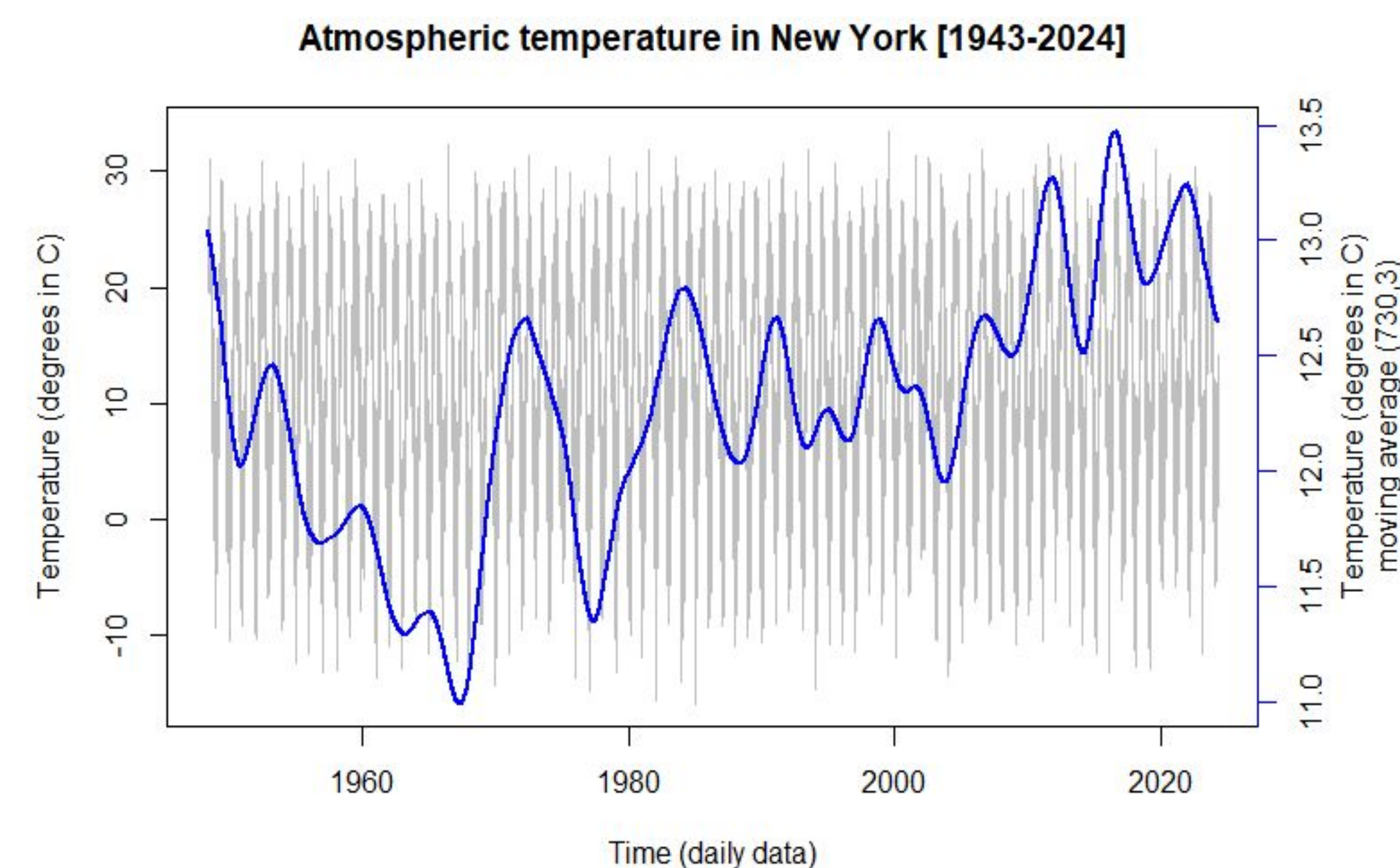


Figure 1: Atmospheric temperature over the past 76 years on Long Island, New York. All data points are shown in gray and represented by the left axis, while average daily temperature is represented by the blue line on the right axis.

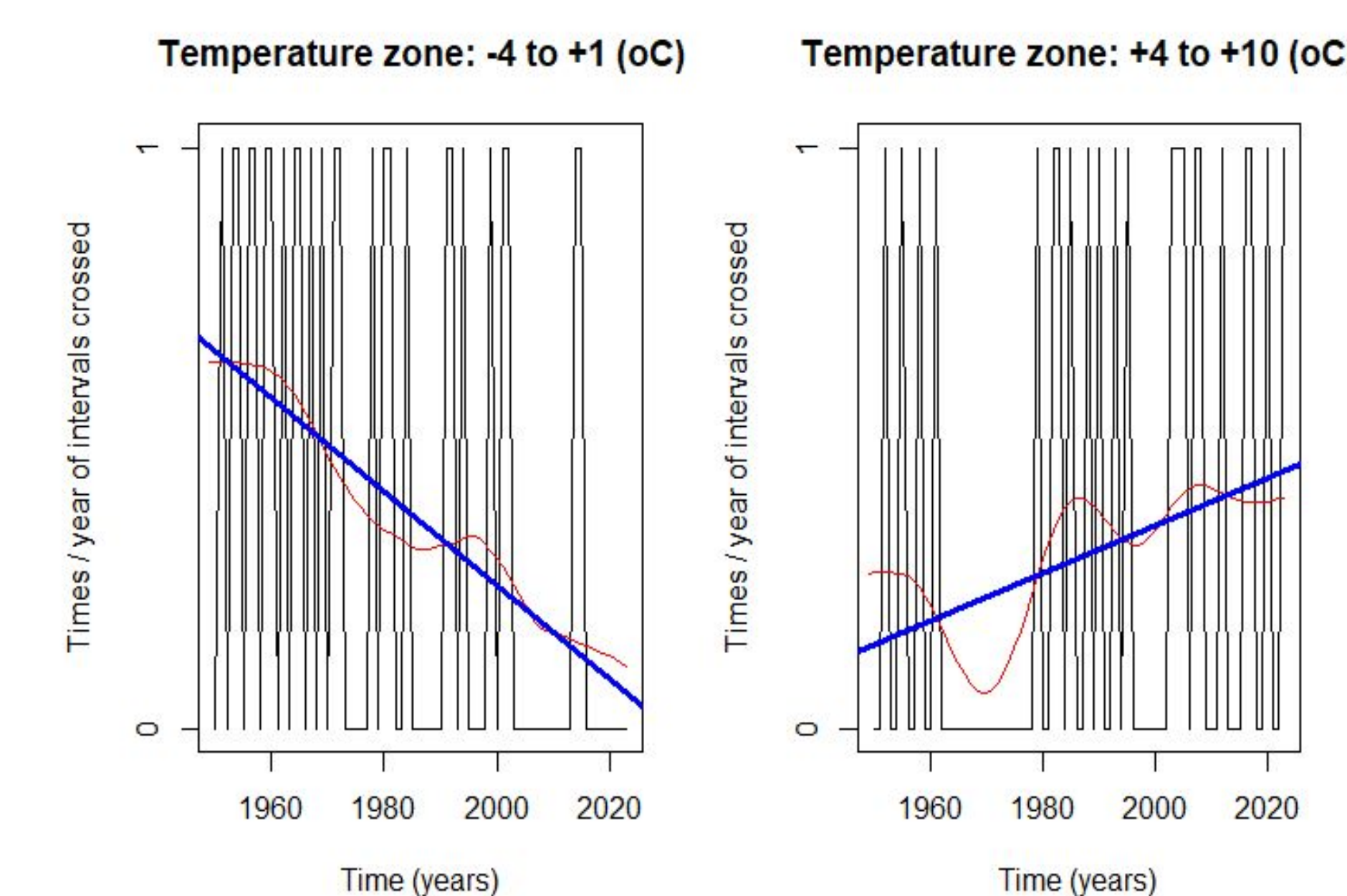
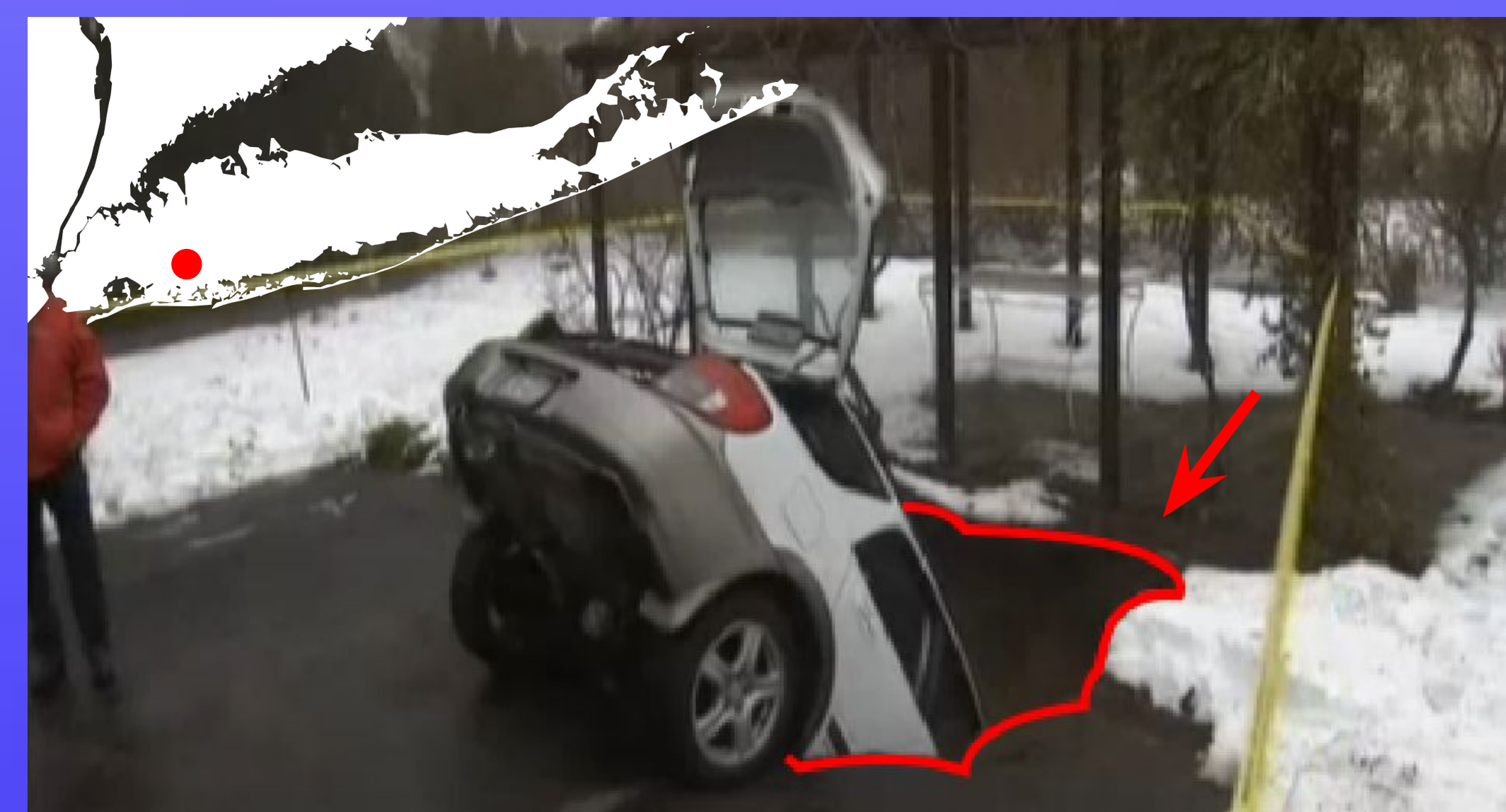


Figure 2: Average times per year atmospheric temperature crossed intervals the -4 to +1°C and +4 to 10°C. The blue line is a linear best fit of the red line, which is a moving average utilizing a KZ filter with 11 years window applied for an iteration of 3 times.



Rockville Sinkhole (February 21, 2014)



Coram Sinkhole (March 1, 2019)

```
5 library(quantreg)
6 library(GSODR)
7 library(kza)
8 library(lubridate)
9
10 # Retrieve data - sometimes it does not download so I save them in hard
11 # file
12 data2 <- GSODR::get_GSOD(years = 1973:2024, station = "744865-14719") # 7
13 data3 <- GSODR::get_GSOD(years = 1948:1972, station = "999999-94789")
14 data <- rbind(data2, data3)
15 # Select only columns of YEARMODA and TEMP
16 data <- data[,c("YEARMODA", "TEMP")]
17 # Rename columns:
18 names(data) <- c("time", "temperature")
19
20 if(!dir.exists("C:\\temp2024\\")){dir.create("C:\\temp2024\\")}
21 saveRDS(data, "C:\\temp2024\\data.R")
22 data <- readRDS("C:\\temp2024\\data.R")
```

R language code showing direct acquisition of temperature data from station

```
24 par(mfrow=c(1,1)) # Set up parameters for graph
25 par(oma=c(1,1,1,4))
26 plot(data$time, data$temperature, pch='.',
27      main="Atmospheric temperature in New York city [1943-2024]",
28      xlab = "Time (daily data)",
29      ylab = "Temperature (degrees in C)", type='l',
30      col = 'grey75')
31
32 par(new=T)
33 kz_temperature <- kz(data$temperature, m = 730, k = 3)
34 plot(data$time, kz_temperature, col='blue', type='l', axes = F, lwd=2, xlab=
35      "Time (daily data)", ylab = "Temperature (degrees in C)", col = 'blue', col.tick
36      mtext(text = paste0("Temperature (degrees in C)/moving average (730,3)"),
37      # text(x = 2000, y = 12, labels = paste0("Temperature (degrees in C)/moving average (730,3)"))
38
39 range(data$time)
40 # Divide data in to years at four different equal intervals
41 temperature <- data$temperature # all data
42 dataz <- data
43
44 range(dataz$temperature)
45 dataz$time
46 dataz$year <- year(dataz$time)
47 dataz$month <- month(dataz$time)
48
```

R language code for the moving average of Figure 1